AQUAFAB INDUSTRIES

Analysing and Exporting Results

Comparing treatments, recomputing from chosen cycles, and exporting what you publish.

For Aquafab Workbench 0.1.0

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Screenshots show simulated machines and modelled example data, not measurements. Aquafab Workbench is not a safety system.

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1 Analysing and exporting results

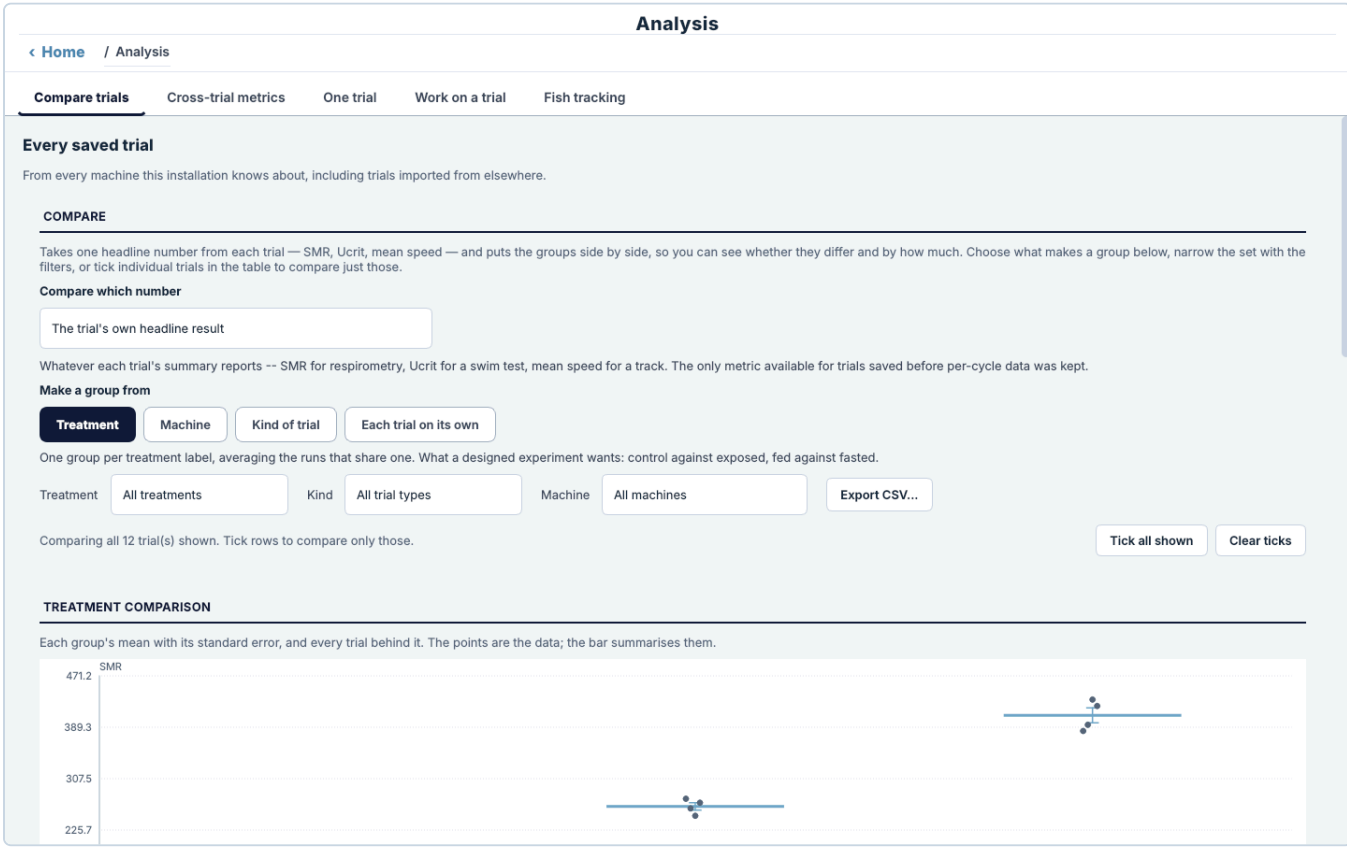
Analysis turns saved trials into the numbers you publish. It needs a license. It is in two places, which are the same view:

- the **Analysis** card on the Launcher: every trial from every machine on this computer, including imported ones;
- a machine's **Analysis** section: that machine's trials only.

> **Analysis never changes a trial.** Each trial keeps the result it reported when saved. Figures recomputed here are shown beside it, and say so.

> The screenshots show modelled example data made for this guide, not measurements.

1.1 Compare treatments



Compare trials.

Compare trials takes one number from each trial and puts groups side by side.

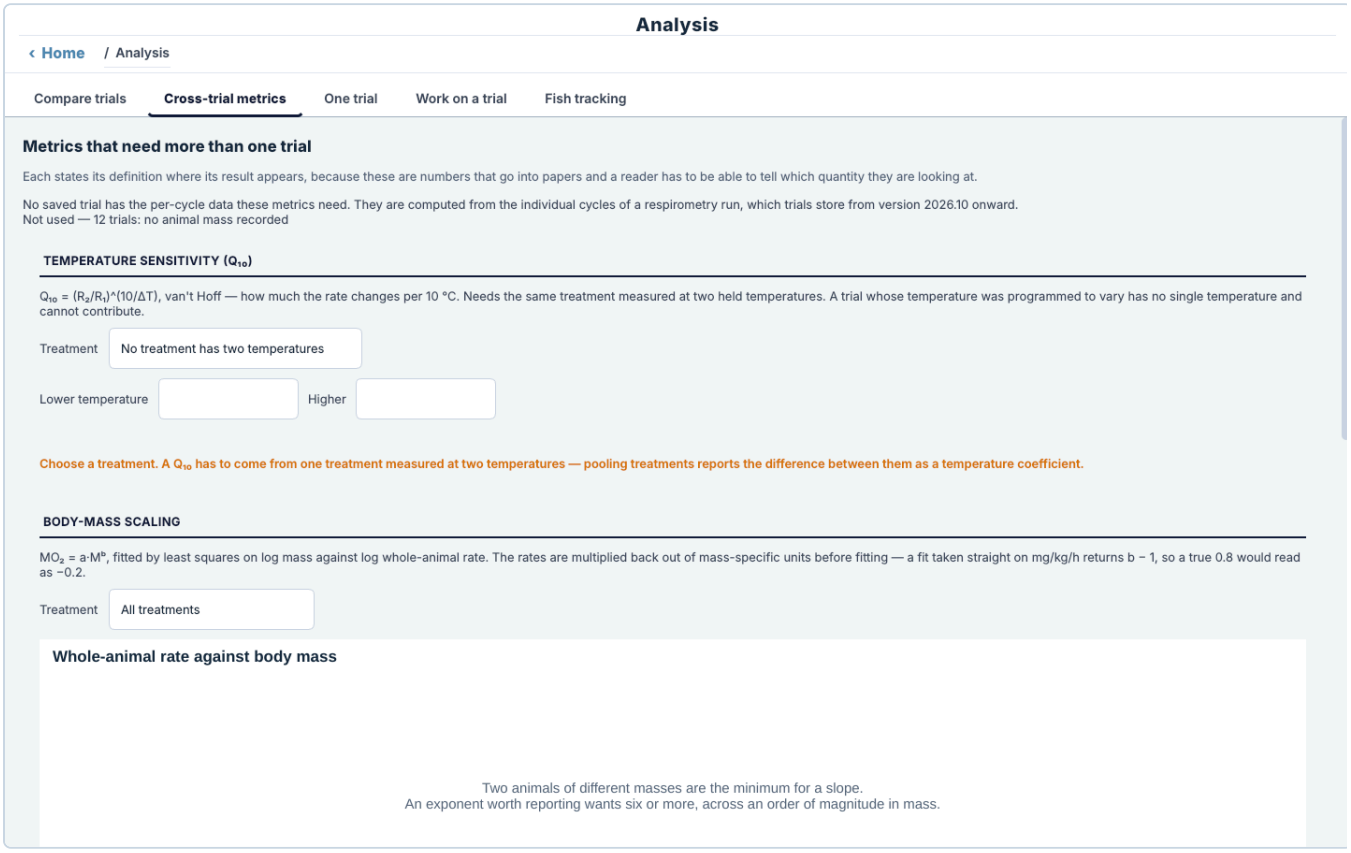
1. **Compare which number**: *The trial's own headline result* (SMR, U_{crit} or mean speed), or, for respirometry trials, a specific metric: SMR by the trial's method or re-estimated, MMR, aerobic scope (absolute or factorial), mean or median MO_2, CV, whole-animal SMR, mass, oxygen levels, mean R^2, accepted and rejected cycles. The definition is shown under the menu.
2. **Make a group from**: **Treatment** (the usual choice), **Machine**, **Kind of trial**, or **Each trial on its own**.
3. Narrow with the treatment, **Kind** and **Machine** filters, or tick trials in the **Trials** table to compare only those.

What you get:

- **Treatment comparison**: each group's mean ± SE, with every trial as a point.
- **Traces, overlaid**: one line per trial (MO_2 or oxygen against cycle, MO_2 against oxygen, or fit quality). Never averaged together.
- **Group statistics**: n, mean, SD, SE and range. SD is how variable the animals were; SE is how well the mean is known. Compare treatments by SE.
- **Every parameter, every trial**: one row per trial, one column per parameter. A blank means it could not be computed, not zero.

Export CSV... writes the visible trials, one row each.

1.2 Cross-trial metrics



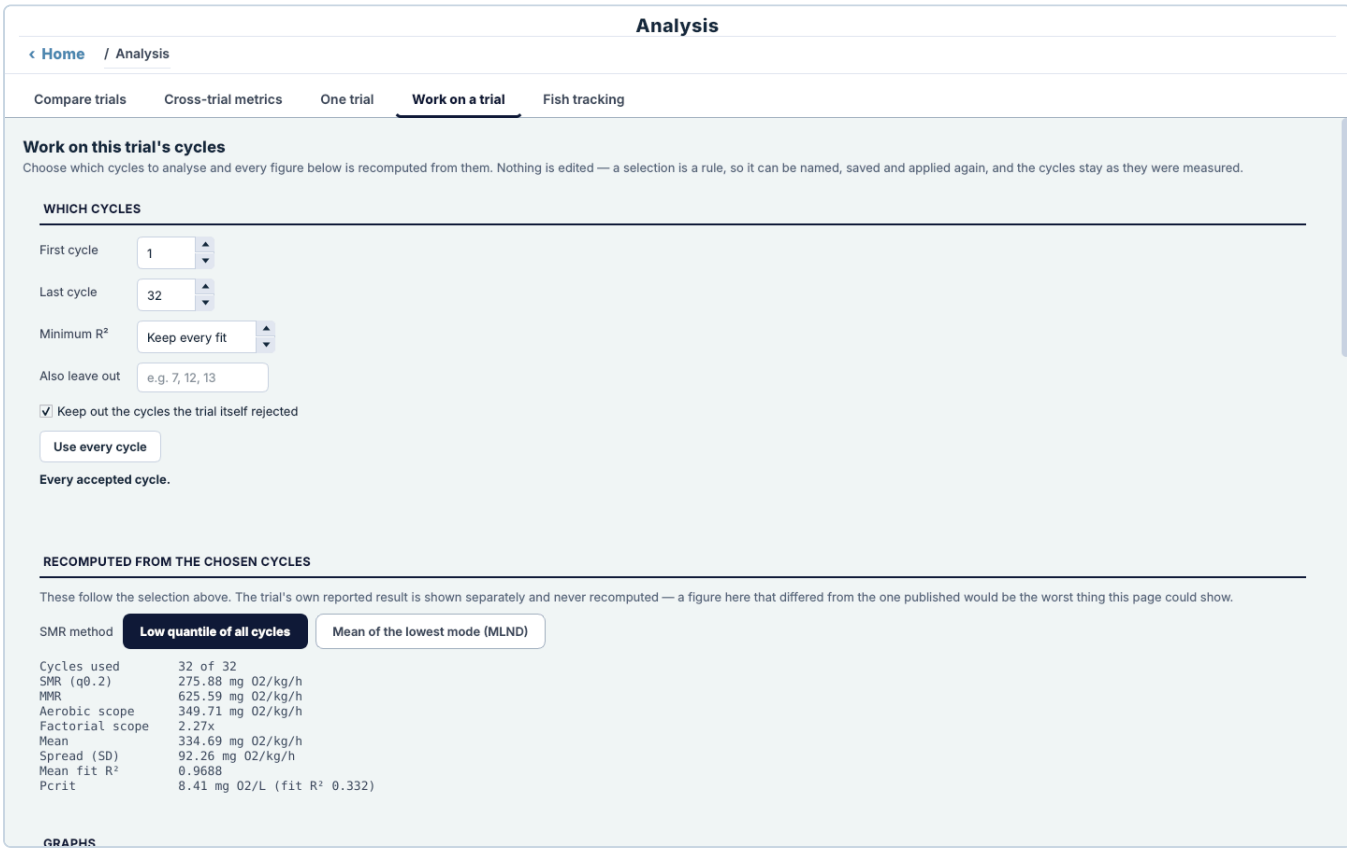
Cross-trial metrics.

Metric	Needs
Q_{10}, $(R_2/R_1)^{(10/\Delta T)}$	One treatment measured at two held temperatures. Choose the treatment and the two temperatures.
Body-mass scaling, $MO_2 = a \cdot M^b$	At least two animals of different masses. Fitted on whole-animal rates, so b is the true exponent.
What the SMR estimate came from	A histogram of every accepted cycle, with the SMR marked. A disturbed animal shows as a high tail.
Oxygen at which aerobic scope has halved	A closed-chamber run where oxygen falls through the trial.

1.3 Look at one trial

One trial (or **Inspect** in Trial History) shows everything about a run: the reported result exactly as saved, every measurement cycle with its MO_2, slope, R^2 and whether it was used, every setting the trial ran with, and notes. **Export CSV...** writes its cycles; **Reopen settings** loads its setup to run again.

1.4 Re-analyse with chosen cycles



Work on a trial.

Work on a trial recomputes everything from the cycles you choose. The trial itself is not edited.

1. Choose the trial.
2. **First cycle**: raise it past the cycles where the animal was settling. **Last cycle**, **Minimum R^2**, and **Also leave out** (for example *7, 12*) remove others. **Keep out the cycles the trial itself rejected** is on by default.
3. Choose the **SMR method**: *Low quantile* or *MLND*.
4. Read the recomputed SMR, MMR, scope, spread and, where oxygen declined, P_{crit}. In the graphs, hollow points are the cycles left out; hover for why.
5. Name the selection (*Settled cycles only*) and press **Save this selection**, so the figure can be reproduced. **Apply selected** brings one back.

1.5 Everything you can export

What	Where
A trial's cycles (CSV)	**Export CSV…** on the trial page, or on the Respirometry Live view with the full oxygen trace
A report (PDF)	**Export PDF…** on Respirometry Live, or **Export PDF…** on Swim Performance
A comparison (CSV)	**Export CSV…** on Compare trials
A chart's data (CSV)	**Export…** on any chart's toolbar: the visible window and series
A track (CSV)	**Export Track…** on Fish Tracking
A vector field (CSV)	**Export Vector Field…** on Flow Field (PIV)

1.6 Methods to cite

SMR: Chabot, Steffensen & Farrell (2016), *J. Fish Biol.* 88, 81–121. Background correction and intermittent flow: Svendsen, Bushnell & Steffensen (2016), *J. Fish Biol.* 88, 26–50. U_{crit}: Brett (1964), *J. Fish. Res. Bd. Canada* 21, 1183–1226. Oxygen solubility: García & Gordon (1992), *Limnol. Oceanogr.* 37, 1307–1312.